
```
clc;
clear all;
close all;

x = rand(1,4);    % 1x4 random numbers
disp('random variable X:')
disp(x)

% Mean using built in MATLAB function
x_m_fun = mean(x);
disp('mean using mean():')
disp(x_m_fun)

% Using formula manually
x_m_calc = sum(sum(x)) / length(x);
disp('mean using formula:')
disp(x_m_calc)

% Using built in MATLAB function
x_var_fun = var(x,1); %----Normalization factor for calculating variance is
N(population of data points)
disp('variance using var():')
disp(x_var_fun)

% Using formula manually:  $\text{Var}(X) = \text{mean}(X^2) - (\text{mean}(X))^2$ 
temp1 = mean(x .* x);    % mean of squares
temp2 = (mean(x))^2;      % square of mean
x_var_calc = temp1 - temp2;
disp('variance using formula:')
disp(x_var_calc)

random variable X:
    0.1656    0.6020    0.2630    0.6541

mean using mean():
    0.4212

mean using formula:
    0.4212

variance using var():
    0.0443

variance using formula:
    0.0443
```

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